GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

Followup 26-18

Acute on chronic hypercapnic respiratory failure in COPD

Patient	A 72-year-old male with centrilobular emphysema prescribed 2 L of home oxygen, treated small cell lung cancer, a prior stroke, carotid stenosis, chronic pain, and kyphosis severe enough that his chin rests on his chest. A prednisone taper had recently been prescribed. **His home oxygen and BiPAP had been returned to the supplier 18 days earlier.**
Outcome	Survived. **The hospital diagnosed acute on chronic hypoxic and hypercapnic respiratory failure from a COPD exacerbation, with suspected pneumonia.** He was intubated in the emergency department and extubated on hospital day 1.
Disposition	Discharged home on hospital day 2, with home oxygen restarted at 3 L.

The call

The call was dispatched as a sick person, an older man who was weak and having trouble breathing. A paramedic from another agency arrived first, found a saturation of 50 percent on room air, placed a non-rebreather, and noted pinpoint pupils. Naloxone was given at that paramedic's request. His son told the crew the oxygen had been taken away 18 days earlier.

The narrative describes him as awake and answering orientation questions, and able to walk from his chair to the stretcher. In the ambulance his breath sounds were diminished bilaterally with expiratory wheezes, and he was pale and cool. His saturation fell whenever his chin dropped to his chest, and the crew repositioned his head repeatedly through transport. Albuterol and then albuterol with ipratropium improved the wheezing.

Times below are from patient contact.

From contact	Finding or action
Before contact	SpO2 50% on room air; non-rebreather; naloxone given
+1 min	Oxygen 4 LPM by nasal cannula
+8 min	Rapid, shallow respirations; diminished breath sounds with expiratory wheezes; pale and cool
+16 min	BP 156/82, pulse 96, SpO2 56%, glucose 126, GCS 13
+19 min	Respirations 28, shallow; SpO2 82%; end-tidal CO2 47 mmHg
+21 min	Albuterol 3 mg by nebulizer
+24 min	SpO2 99%; end-tidal CO2 43 mmHg
+26 min	IV, right forearm
+33 min	Albuterol with ipratropium by nebulizer
+37 min	Leaves scene

From contact	Finding or action
+62 min	Arrives at the emergency department

The numbers to hold onto are a saturation of 50 percent before contact, an end-tidal CO2 in the 40s, and an arterial PCO2 of 66 an hour later.

Primary impression

The crew's impression was shortness of breath. The hospital named it acute on chronic hypoxic and hypercapnic respiratory failure, from a COPD exacerbation in a man who no longer had the home oxygen and BiPAP that had been keeping his chronic respiratory failure in balance. The crew recognized that his saturation depended on the position of his head and kept correcting it through transport, which is the airway management his anatomy required.

Hospital course

Emergency department, day 0. He arrived minimally responsive, with a GCS of 6 and a saturation of 53 percent. He was placed on a non-rebreather and intubated on the first attempt by video laryngoscopy with etomidate and rocuronium. His arterial gas before intubation showed a pH of 7.26, a PCO2 of 66, and a bicarbonate of 29.1. His chest x-ray showed mild basilar atelectasis or infiltrates, worse on the right. He received methylprednisolone, ceftriaxone and bronchodilators, and was admitted to intensive care on propofol.

Hospital day 1. He was extubated in the morning and moved out of intensive care. He tolerated BiPAP for about an hour afterward.

Hospital day 2. He was eating, walking, and said he felt great. His walk test showed a saturation of 88 percent at rest on room air, 83 percent with exertion, and 91 percent with exertion on 3 L. He was discharged home with oxygen at 3 L, his inhalers, his scheduled azithromycin, and three more days of amoxicillin-clavulanate for suspected pneumonia. He agreed to take the oxygen again.

Final diagnoses: Acute on chronic hypoxic and hypercapnic respiratory failure. COPD exacerbation. Suspected pneumonia. Acute encephalopathy. Hyponatremia.

Education points

Chronic hypercapnic respiratory failure

Let's start with what chronic hypercapnia is. In advanced emphysema the lungs cannot move enough air to clear the carbon dioxide the body produces, and arterial CO2 sits above normal all the time. The kidneys compensate over days by holding bicarbonate, which pulls the pH back toward normal. A patient like this lives with a PCO2 in the 50s and a bicarbonate near 30, and feels fine.

That balance has no reserve. Home oxygen keeps his saturation high enough that his heart and brain are not hypoxic. BiPAP at night does part of the work of breathing while he sleeps, when ventilation falls, and keeps the CO2 from climbing.

When carbon dioxide rises faster than the kidneys can compensate, the pH falls. His first gas showed a pH of 7.26 with a PCO2 of 66 and a bicarbonate of 29.1. The bicarbonate says the hypercapnia was old. The pH says it had

recently become worse. Rising CO2 sedates, and drowsiness reduces breathing further. This is called CO2 narcosis, and its course is a falling level of consciousness that can be gradual or abrupt.

These patients can deteriorate within minutes. With his chronic hypoxemia, his oxygen tension sat on the steep part of the oxyhemoglobin dissociation curve, where a small fall in oxygen tension produces a large fall in saturation. Taking the oxygen off for a move, a change of monitor or a change of position can drop the saturation sharply within a minute or two. Carbon dioxide has a tipping point too: once drowsiness reduces breathing, the rise accelerates, and a patient who was talking can be obtunded a short time later. The transfer at the destination, with its moves, equipment changes and repositioning, is one of the moments when this happens.

Oxygen in a patient with chronic hypercapnia is a balance. A saturation of 50 percent kills faster than carbon dioxide does, and correcting severe hypoxemia comes first. Once the saturation is up, the target is 88 to 92 percent, because more oxygen than that can raise carbon dioxide by releasing hypoxic vasoconstriction in poorly ventilated lung and by blunting the drive to breathe.

End-tidal CO2 in emphysema

End-tidal CO2 normally reads a few mmHg below arterial CO2. In emphysema the gap is much wider. Destroyed alveoli and their lost capillaries create dead space: lung that is ventilated but not perfused. Air from those regions carries no carbon dioxide, and it dilutes the exhaled sample.

In a prospective study of patients with COPD exacerbations in the emergency department, end-tidal CO2 by sidestream sampling read on average about 13 mmHg below arterial CO2 before treatment, and in some patients the gap was over 30. Shallow breathing widens it further, because each breath is mostly dead space, and a nasal cannula sample is diluted by mouth breathing and by oxygen flow.

His end-tidal readings were 47 and 43 early in transport, and his arterial PCO2 was 66 an hour later. In a patient with emphysema, a normal end-tidal number does not mean a normal arterial CO2. A high end-tidal number means the arterial CO2 is at least that high. The waveform and the trend, together with the patient's level of consciousness, carry more information than any single value.

Kyphosis and the airway

Severe thoracic kyphosis compresses the chest and stiffens it, which reduces lung volume and adds to the work of breathing in a patient who already cannot ventilate well. It also brings the chin toward the chest. Neck flexion narrows the upper airway, and in a drowsy patient it can close the airway entirely.

His saturation fell whenever his chin touched his chest. The fix is position: support the head in a neutral position with padding behind it rather than forcing the neck back, which the kyphosis may not allow, and recheck every time he moves or drifts.

> **Further reading on COPD exacerbation.** The EMCrit Internet Book of Critical Care chapter on COPD, including chronic hypercapnia, oxygen targets, and when BiPAP is needed. https://emcrit.org/ibcc/copd/
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> **Further reading on end-tidal CO2 in COPD.** The emergency department study comparing sidestream end-tidal CO2 with arterial CO2 in COPD exacerbations, with the size and spread of the gap. https://pmc.ncbi.nlm.nih.gov/articles/PMC10550360/

Prehospital considerations

Ask about home oxygen and BiPAP, and whether they are still being used. A patient who has stopped either has lost support that was keeping chronic respiratory failure stable. The answer belongs in the handoff.

In emphysema, end-tidal CO2 underestimates arterial CO2. Read it as a floor, follow the trend and the waveform, and treat a falling level of consciousness as rising carbon dioxide until proven otherwise.

In severe kyphosis, head position is airway position. This crew recognized that his saturation followed his chin and corrected it through transport.

The interventions

PREHOSPITAL INTERVENTIONS

Oxygen. A non-rebreather for a saturation of 50 percent, then a nasal cannula, and oxygen through the nebulizer during treatments.

Naloxone. Given for pinpoint pupils and depressed breathing. The hospital recorded that the response was minimal.

Albuterol 3 mg, then albuterol with ipratropium, by nebulizer. Albuterol relaxes bronchial smooth muscle through beta-2 receptors. Ipratropium blocks the muscarinic receptors that tighten the airways in COPD, and adds to the albuterol's effect. His wheezing improved.

End-tidal CO2 monitoring. Applied in the ambulance and trended through transport.

Head repositioning. Repeated through transport as his saturation fell with neck flexion.

IN-HOSPITAL INTERVENTIONS

Rapid sequence intubation. First attempt by video laryngoscopy with etomidate and rocuronium, for a GCS of 6 and hypercapnic respiratory failure.

Methylprednisolone, then prednisone. Corticosteroids shorten a COPD exacerbation and reduce treatment failure.

Ceftriaxone, then amoxicillin-clavulanate, with azithromycin. For suspected pneumonia. The azithromycin was already part of his home regimen, taken three days a week to reduce exacerbations.

Home oxygen. Reordered at 3 L after a walk test, with his agreement.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Blood gases

Lab	Before intubation	On the ventilator	Reference range
pH, arterial	7.26	Normal	7.35 to 7.45
PCO2, arterial (mmHg)	66	57	35 to 48
PO2, arterial (mmHg)	109	48	83 to 108
Bicarbonate, arterial (mmol/L)	29.1	31.5	21.0 to 28.0
O2 saturation, arterial (%)	Normal	80	94 to 98

A PCO2 of 66 with a bicarbonate of 29.1 is acute on chronic respiratory acidosis. The bicarbonate is the kidneys' compensation for chronic hypercapnia, and a pH of 7.26 means the CO2 had climbed faster than that compensation could follow. On the ventilator his PCO2 fell to 57, which is closer to his likely baseline. His oxygen tension of 48 on the ventilator shows how much lung was not exchanging gas, and why his saturation could not be held up on scene.

Heart, chemistry and blood count

Lab	Value	Reference range
Pro-BNP (ng/L)	2,299	125 or less
hs troponin T, baseline (ng/L)	28	23 or less
hs troponin T, 2 hour (ng/L)	30	Less than 23
Sodium (mmol/L)	130	136 to 145
Hemoglobin (g/dL)	11.4	14.0 to 18.0
Red cell distribution width (%)	20.3	11.0 to 16.0

His troponin was flat, which is strain from hypoxemia rather than an infarction. His lactate was normal and a respiratory pathogen panel was negative.

A positive fentanyl screen after a dose of hospital fentanyl proves nothing. His urine screen was presumptively positive for fentanyl and cannabinoids, and negative for opiates and oxycodone. He received fentanyl 100 mcg IV after intubation, and the record does not show whether the urine was collected before or after. No opioid is on his medication list. Before reading a screen as evidence of an ingestion, check what the patient received before the sample was taken.

Learning points

1. **A raised bicarbonate in a hypercapnic patient means the carbon dioxide has been high for days.** His was 29.1 with a PCO2 of 66 and a pH of 7.26: chronic hypercapnia, recently worse.
2. **Stopping home oxygen and BiPAP removes the support holding chronic respiratory failure in balance.** Both had been returned 18 days earlier. Ask whether the equipment is still in use.
3. **In emphysema, end-tidal CO2 reads well below arterial CO2.** In emergency department patients with COPD the gap averaged about 13 mmHg and sometimes exceeded 30. Read the number as a floor and follow the trend.
4. **A patient with chronic hypercapnia and no reserve can go from talking to obtunded within minutes.** Every move and every interruption of oxygen, including the transfer at the destination, is a point of risk. Treat severe hypoxemia first, then aim for a saturation of 88 to 92 percent.
5. **In severe kyphosis, head position is airway position.** His saturation fell every time his chin reached his chest, and the crew corrected it through transport.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at https://www.berrienmca.org/followup/